1. Prerequisites	A knowledge of the specification for P1 and its associated formulae is assumed and may be tested.
2. Examination	• First assessment: June 2019. • The assessment is 1 hour and 30 minutes. • The assessment is out of 75 marks. • Students must answer all questions. • Calculators may be used in the examination. Please see *Appendix 6: Use of calculators*. • The booklet *Mathematical Formulae and Statistical Tables* will be provided for use in the assessments.
2. Notation and formulae	Students will be expected to understand the symbols outlined in *Appendix 7: Notation*.

Formulae that students are expected to know are given below and will **not** appear in the booklet; *Mathematical Formulae and Statistical Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.

Laws of logarithms

$\log_a x + \log_a y \equiv \log_a (xy)$

$\log_a x - \log_a y \equiv \log_a \left(\frac{x}{y}\right)$

$k \log_a x \equiv \log_a (x^k)$

Trigonometry

$\sin^2 A + \cos^2 A \equiv 1$

$\tan\theta \equiv \frac{\sin\theta}{\cos\theta}$

Area

P2.3 Unit content

What students need to learn:		Guidance
1. Proof		
1.1	Understand and use the structure of mathematical proof, proceeding from given assumptions through a series of logical steps to a conclusion; use methods of proof stated below:	
1.2	Proof by exhaustion	Proof by exhaustion. This involves trying all the options. Suppose x and y are odd integers less than 7. Prove that their sum is divisible by 2.
1.3	Disproof by counter example.	Disproof by counter example – show that the statement "$n^2 - n + 1$ is a prime number for all values of n" is untrue.